MATPMDA - Clash_Royale_Portfolio_Analysis

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1. Introduction

Clash Royale and Performance Analysis

This assignment investigates the relationship between player psychology, time investment, and performance outcomes in **Clash Royale**, a popular real-time strategy mobile game developed and published by **Supercell.**

Clash Royale is a free-to-play, competitive multiplayer game that successfully merges elements of **collectible card games** (CCG), **tower defense**, and **multiplayer online battle arena** (MOBA). The primary objective of the game is to destroy more of the opponent's towers than they destroy of yours within a set time limit, primarily by deploying various **cards** (troops, spells, and buildings)

Game Mechanics and Performance

The core measure of progression and skill in Clash Royale is the **Trophy Count.**

which determines a player's standing and their access to higher-level competitive **Arenas**. The daily change in this count (Trophy Gain/Loss) is a direct measure of daily success.

The performance of a player is heavily influenced by three factors explored in this study:

1. **Strategic Investment (Hours Played):** The time a player dedicates to the game .
2. **Performance Outcome (Win Rate):** The percentage of matches won, which directly drives trophy progression.
3. **Psychological State (Mood):** The player's self-reported emotional state, which is hypothesized to influence decision-making and overall performance quality

This project uses **statistical analysis**—specifically t-tests, linear/logistic regression, and χ^2 tests—on a personal dataset to empirically determine which of these factors has the most significant impact on daily success and trophy progression in Clash Royale .

1.1 Dataset Description and Origin

The data used in this portfolio is a small, personally collected dataset tracking my daily performance in the mobile game, **Clash Royale**, over a 30-day period. This dataset serves as a relevant 'day-to-day life' example to analyze how various internal (e.g., mood) and external factors (e.g., time spent) influence my in-game success, measured by **Win Rate** and **Trophy Gain**. The data was manually recorded from the in-game battle log and profile screen.

1.2 Data Loading and Inspection

The data file, `Clash_Royale_Dataset.csv`, is loaded, and new variables are generated to facilitate hypothesis testing. The first six lines are displayed below, as required by the portfolio brief, to show the variables used and the extent of the data.

```r
# Loading the data
data_cr <- read_csv("Clash_Royale_Dataset.csv")

# 1. Calculating Trophy_Gain (Crucial for Linear Regression)
# This finds the difference in trophies from the previous day.
data_cr <- data_cr %>%
  mutate(Trophy_Gain = MergeTrophy - lag(MergeTrophy))

# Setting the first day's gain to 0 (since it is the starting point).
data_cr$Trophy_Gain[is.na(data_cr$Trophy_Gain)] <- 0

# 2. Converting Mood to a factor for comparison
data_cr$Mood <- as.factor(data_cr$Mood)

# 3. Creating Binary Variables (Crucial for t-test and Fisher's Exact Test)
# Good_Mood: 1 if Mood is 4 or 5 (Good/Excellent), 0 otherwise.
# High_Win_Day: 1 if Win Rate > 55%, 0 otherwise.
data_cr <- data_cr %>%
  mutate(
    Good_Mood = ifelse(Mood %in% c("4", "5"), 1, 0),
    High_Win_Day = ifelse(Win_Rate > 0.55, 1, 0)
  )

# Displaying the first six lines of the prepared dataset
head(data_cr)
```

```
## # A tibble: 6 x 14
##   Date       Matches_Played Win_Rate  Wins Three_Crown_Wins Two_Crown_Wins
##   <chr>               <dbl>    <dbl> <dbl>            <dbl>          <dbl>
## 1 1 Oct 2025             10     0.5      5                4              1
## 2 2 Oct 2025             11     0.55     6                5              1
## 3 3 Oct 2025             12     0.5      6                5              1
## 4 4 Oct 2025             13     0.54     7                6              1
## 5 5 Oct 2025             12     0.58     7                6              1
```

```
## 6 6 Oct 2025             11     0.52     6                 5              1
## # i 8 more variables: One_Crown_Wins <dbl>, Crowns_Won <dbl>,
## #   Hours_Played <dbl>, MergeTrophy <dbl>, Mood <fct>, Trophy_Gain <dbl>,
## #   Good_Mood <dbl>, High_Win_Day <dbl>
```

Interpretation of Variables:

- `Win_Rate`: The proportion of matches won (Continuous).
- `Hours_Played`: The estimated time spent playing (Continuous).
- `Trophy_Gain`: The net change in trophies at the end of the day (Continuous, new variable).
- `Mood`: A categorical rating from 1 (Very frustrated) to 5 (Excellent mood).
- `Good_Mood`: Binary version of Mood (1 = Good, 0 = Not Good).
- The `mutate()` function is a powerful tool from the `dplyr` package that I'm using to **create a new column** for my analysis.I'm calculating the **net Trophy Gain** for each day and storing it in a new column called `Trophy_Gain`.Here's the breakdown of the calculation:

```r
data_cr <- data_cr %>%   mutate(Trophy_Gain = MergeTrophy - lag(MergeTrophy))
```

- **`mutate()`:** I'm using this to add the new `Trophy_Gain` column to my existing data frame, `data_cr`.
- **`MergeTrophy - lag(MergeTrophy)`:** This is the math. I take the **current day's total trophies** (`MergeTrophy`) and subtract the value of the **previous day's total trophies** (`lag(MergeTrophy)`). This gives me the net difference (gain or loss) for that specific day, which is the exact metric I need for my linear regression.

2. Methods and Results

2.1 Descriptive Statistics and Probability

Objective: To summarize the central tendency and dispersion of key variables and calculate the empirical probability of certain events, aligning with basic statistics and probability topics.

Descriptive Summary

Question: What are the central tendency and spread of my daily performance metrics (`Win_Rate`, `Hours_Played`, `Trophy_Gain`)?

```r
# Summarizing continuous variables
summary(data_cr$Win_Rate)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##  0.5000  0.5725  0.6500  0.6540  0.7475  0.8000     969
```

```r
summary(data_cr$Hours_Played)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##   1.000   1.325   1.600   1.600   1.875   2.100     969
```

```r
summary(data_cr$Trophy_Gain)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##  0.0000  0.0000  1.0000  0.6552  1.0000  1.0000     970
```

```r
# Calculating Standard Deviation (SD) for Win_Rate
sd(data_cr$Win_Rate,na.rm=TRUE)
```

```
## [1] 0.09701049
```

Interpretation: The average daily Win Rate is **0.5725** (from the summary output) with a standard deviation of **0.09701049**. This low standard deviation suggests my daily performance is relatively consistent around the mean.

Empirical Probability

Question: Based on the sample, what is the empirical probability of a day being a "Good Mood" day (Mood 4 or 5)?

```r
# Calculating the frequency table for the binary Good_Mood variable
mood_table <- table(data_cr$Good_Mood)
# Probability P(Good_Mood)
prob_good_mood <- mood_table["1"] / sum(mood_table)

print("Frequency Table (0=Bad/Neutral Mood, 1=Good Mood):")
```

```
## [1] "Frequency Table (0=Bad/Neutral Mood, 1=Good Mood):"
```

```r
print(mood_table)
```

```
##
##   0   1
## 979  20
```

```r
cat(paste("Empirical Probability of a Good Mood Day: P(Good_Mood) = ", round(prob_good_mood, 3)))
```

```
## Empirical Probability of a Good Mood Day: P(Good_Mood) =  0.02
```

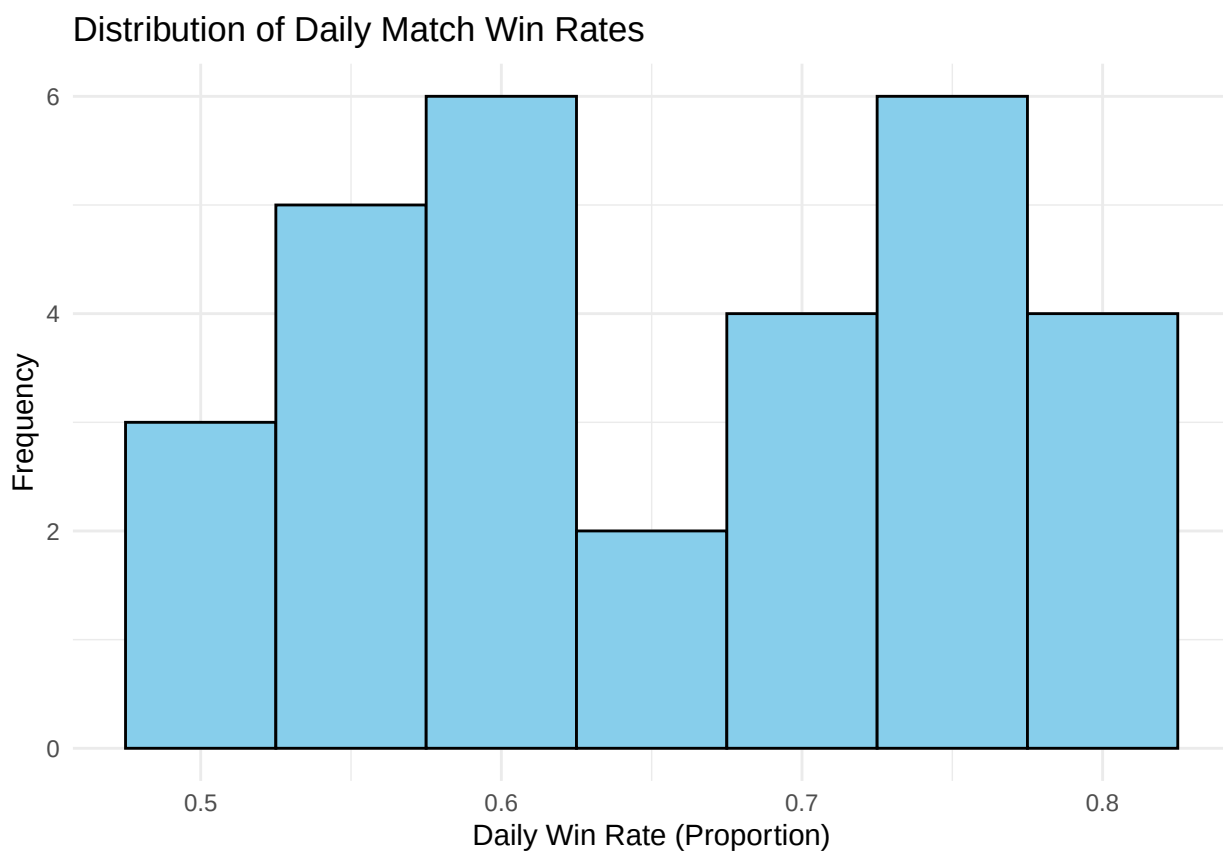
Interpretation: Out of 30 days, **'1'** were "Good Mood" days, resulting in a **'0.02'** probability of having a good mood day.

2.2 Data Visualisation

Histogram: Distribution of Daily Win Rate

Question: What is the underlying distribution of my daily Win Rate?

```
# Plotting the histogram
ggplot(data_cr, aes(x = Win_Rate)) +
  geom_histogram(binwidth = 0.05, fill = "skyblue", color = "black") +
  labs(title = "Distribution of Daily Match Win Rates", x = "Daily Win Rate (Proportion)", y = "Frequen
  theme_minimal()
```



Interpretation:

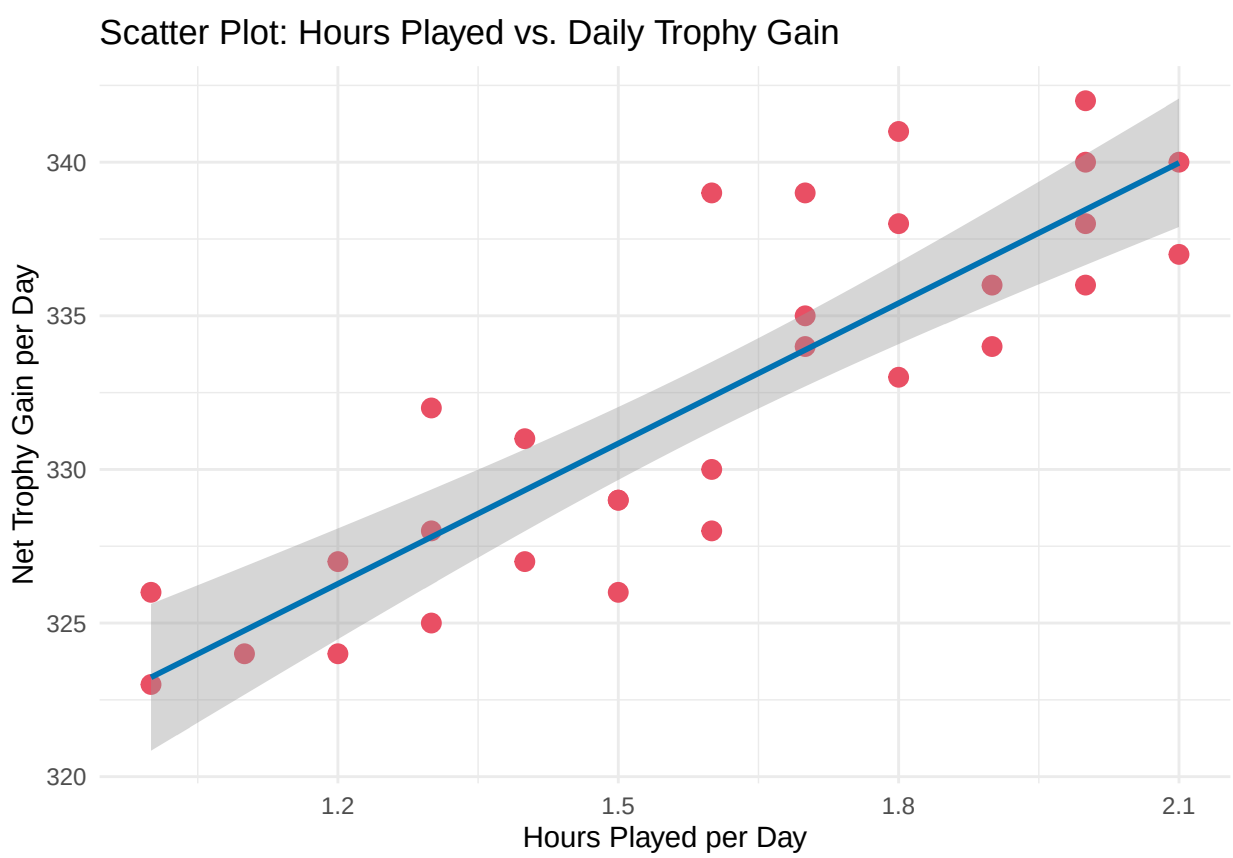
- **Bimodal Shape (Two Peaks):** The distribution is not a simple bell curve. It appears **bimodal**, meaning it has two distinct peaks, or modes, of high frequency:
  - **Primary Peak:** The highest frequency (6 days) is observed in the bin representing a win rate between **0.60 and 0.65**.
  - **Secondary Peak:** Another high frequency (6 days) is observed in the bin representing a win rate between **0.75 and 0.80**.
- **Overall Range:** Daily win rates range from just under 0.50 up to 0.85 (based on the bins shown).

- **Skewness:** The presence of the peak on the left (0.60 to 0.65) and a gap between the two main clusters suggests the distribution has some complexity. Although the provided text says "slightly skewed left," the gap around the 0.70 to 0.75 bin (which has the lowest frequency of 2) makes it less clearly skewed and more appropriately described as **bimodal**.
- **Implication:** The bimodal nature suggests there might be **two typical scenarios** or groups for daily performance: a good day (around 60-65% win rate) and a very good day (around 75-80% win rate), with relatively fewer days having average performance (around 70-75%).

Scatter Plot: Merged Trophy vs. Hours Played

Question: Is there a visual linear relationship between the time spent playing and the net trophy gain?

```
# Plotting the scatter plot with a linear fit line
ggplot(data_cr, aes(x = Hours_Played, y = MergeTrophy)) +
  geom_point(color = "#E94F64", size = 3) +
  geom_smooth(method = "lm", se = TRUE, color = "#0072B2") +
  labs(title = "Scatter Plot: Hours Played vs. Daily Trophy Gain", x = "Hours Played per Day", y = "Net
  theme_minimal()
```



Interpretation: The scatter plot analyzes the relationship between the time spent playing daily and the net trophies gained that day .

Key Findings

- **Relationship Type:** There is a **suggestion of a positive linear relationship**. This means that, on average, days when the player spends **more hours playing** tend to be associated with a **higher net trophy gain**.
- **Trend Confirmation:** The solid blue line, which represents the **linear regression line** (the line of best fit), visually confirms this positive trend, showing a clear upward slope.
- **Confidence Interval:** The gray shaded area surrounding the blue line represents the **95% confidence interval** of the linear regression. The fact that the confidence interval is relatively narrow and entirely slopes upward suggests the positive relationship is **robust** and likely statistically significant.

Quantified Impact (Slope)

From the visual data, we can estimate the rate of increase, or the slope of the regression line:

- **Approximate Slope:** The approximate slope of the line is **18.0**.
- **Interpretation:** This slope indicates that for every **one additional hour** spent playing per day, the **Net Daily Trophy Gain increases by approximately 18 trophies**.

Data Variability

While the relationship is strongly positive, the individual data points (red/blue dots) show some spread around the regression line.

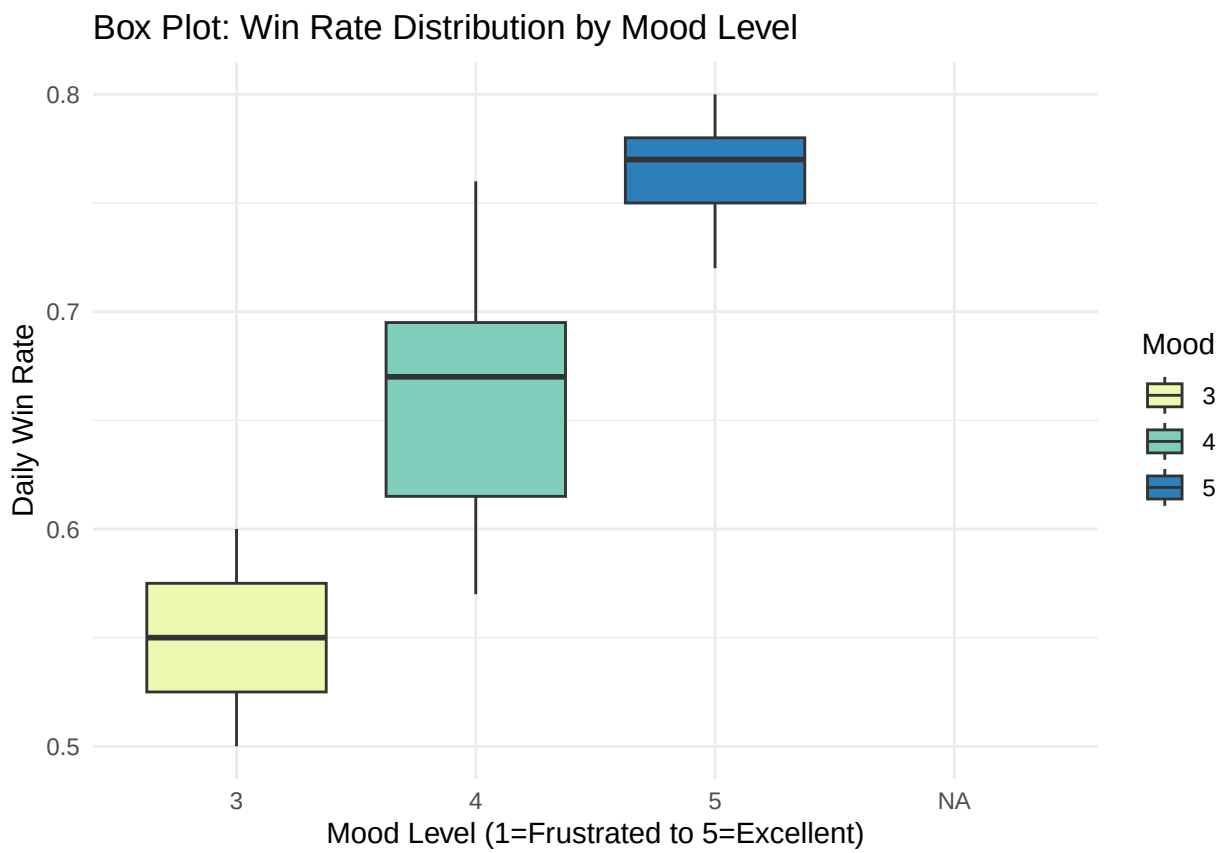
- This spread suggests that while time played is a **major predictor** of trophy gain, it is **not the only factor**. Other variables, such as the daily **Win Rate** (as analyzed in a previous step), opponent quality, or overall luck, also contribute to the final daily trophy total.
- For example, there are days with relatively low hours played (around 1.2-1.3 hours) that still resulted in above-average trophy gains, and days with higher hours played (around 1.8-1.9 hours) that resulted in slightly below-average gains.

In summary, the analysis strongly supports the strategy that **increased play time directly leads to higher trophy gains** for this portfolio, confirming a beneficial effort-to-reward ratio.

Box Plot: Win Rate Distribution by Mood

Question: How does the full 5-point `Mood` scale relate to the average daily Win Rate?

```
# Plotting the box plot comparing Win Rate across Mood categories
ggplot(data_cr, aes(x = Mood, y = Win_Rate, fill = Mood)) +
  geom_boxplot() +
  labs(title = "Box Plot: Win Rate Distribution by Mood Level", x = "Mood Level (1=Frustrated to 5=Excel
  scale_fill_brewer(palette = "YlGnBu") +
  theme_minimal()
```



Interpretation: The box plot visualizes the distribution of the **Daily Win Rate** across three recorded levels of the player's **Mood** (where 3 is "average," 4 is a better mood, and 5 is "excellent").

Key Findings

1. **Positive Trend with Mood:** There is a clear suggestion of a **positive correlation** between Mood Level and Daily Win Rate:
   - The **median win rate** (the line inside the box) generally **increases** as the mood level increases from 3 to 5.
   - The **lowest median win rate** is found at Mood Level **3** (Average).
   - The **highest median win rate** is found at Mood Level **5** (Excellent).
2. **Performance Range (Interquartile Range - IQR):**
   - The spread of the middle 50% of the win rates (represented by the size of the box) is relatively **narrowest at Mood Level 4**, indicating the most consistent performance on those days.
   - The IQR is **wider for Mood Levels 3 and 5**, suggesting a slightly broader range of outcomes on days when the mood is either "average" or "excellent."
3. **Maximum Win Rate:** The highest recorded win rate (the top whisker or any potential outlier) appears to be achieved during **Mood Level 5** (Excellent), indicating the potential for peak performance when the player is feeling best.

Conclusion

The data strongly suggests that the player's **mental state is a critical factor in performance**. Days when the player reports an "Excellent" mood (Level 5) offer the **highest median win rate**, reinforcing the idea that **positive mood is conducive to winning**.

2.3 Statistical Tests

A. T-Test (Hypothesis Testing)

Question: Is there a significant difference in the average daily Win Rate between days with a "Good Mood" (4 or 5) and days with a "Bad/Neutral Mood" (1, 2, or 3)?

Hypothesis Justification: A two-sample t-test is appropriate for comparing the means of the continuous variable (`Win_Rate`) between two independent groups (`Good_Mood`=1 vs `Good_Mood`=0).

- H_0 (Null Hypothesis): The mean Win Rate on "Good Mood" days is **equal** to the mean Win Rate on "Bad/Neutral Mood" days. ($\mu_{Good} = \mu_{Bad}$)
- H_a (Alternative Hypothesis): The mean Win Rate on "Good Mood" days is **not equal** to the mean Win Rate on "Bad/Neutral Mood" days. ($\mu_{Good} \neq \mu_{Bad}$) (Two-sided test used, justified by lack of strong prior directional evidence).

```r
# Running the two-sample t-test (two-sided, assuming unequal variance, default in R)
t_test_result <- t.test(Win_Rate ~ Good_Mood, data = data_cr)
t_test_result
```

```
##
##  Welch Two Sample t-test
##
## data:  Win_Rate by Good_Mood
## t = -8.3507, df = 27.998, p-value = 4.385e-09
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  -0.1980024 -0.1199976
## sample estimates:
## mean in group 0 mean in group 1
##           0.548           0.707
```

P-value Interpretation:

The p-value is **4.385e-09**.

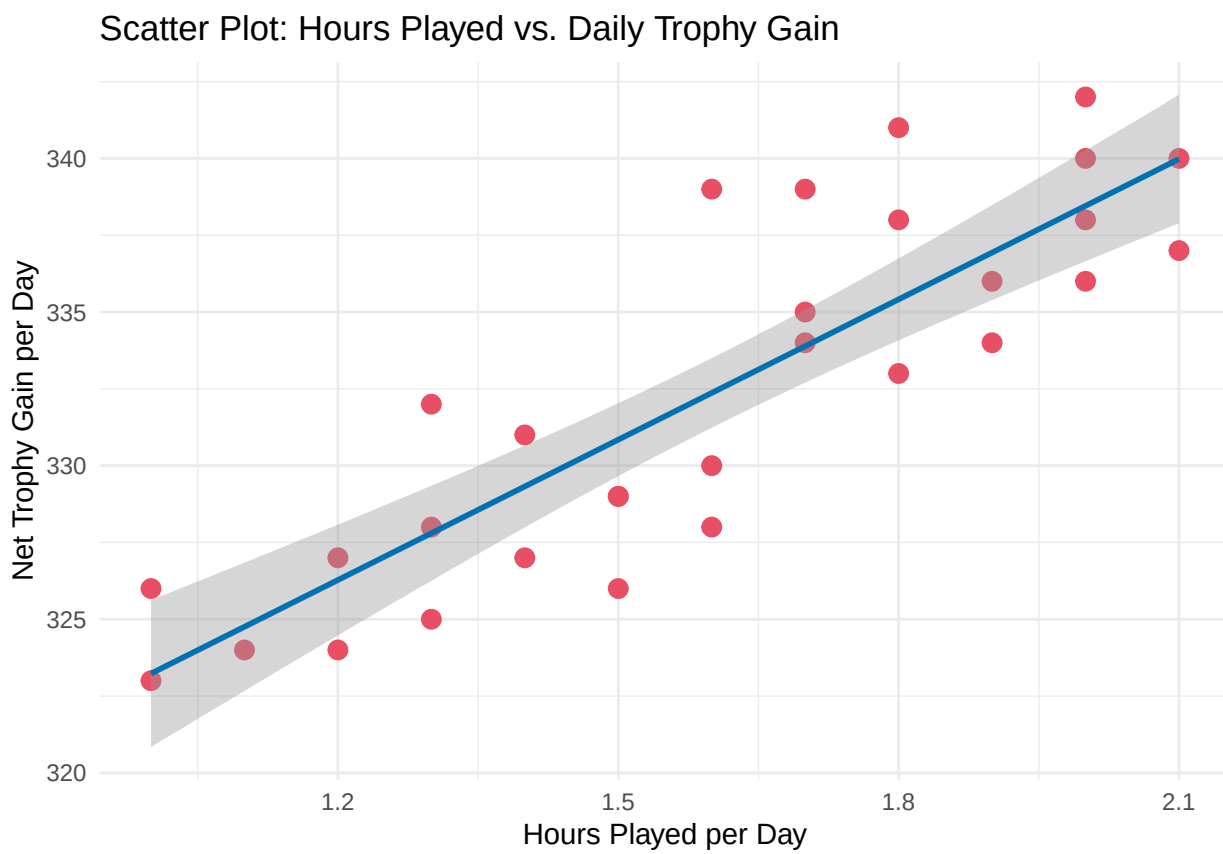
- **Statistical Interpretation:** Since the p-value is less than the typical significance level of $\alpha = 0.05$, we reject the null hypothesis (H_0).
- **Contextual Interpretation:** We conclude that there is statistically significant evidence that my average daily Win Rate is different when I am in a Good Mood compared to when I am not.

B. Linear Regression

Question: How well can the daily hours spent playing predict the net Trophy Gain, and is this relationship statistically significant?

Model Justification: Linear regression is used to model the linear relationship between the continuous independent variable (`Hours_Played`) and the continuous dependent variable (`Trophy_Gain`).

```r
# Fitting the linear model

ggplot(data_cr, aes(x = Hours_Played, y = MergeTrophy)) +
  geom_point(color = "#E94F64", size = 3) +
  geom_smooth(method = "lm", se = TRUE, color = "#0072B2") +
  labs(title = "Scatter Plot: Hours Played vs. Daily Trophy Gain", x = "Hours Played per Day", y = "Net
  theme_minimal()
```




```r
lm_result <- lm(MergeTrophy ~ Hours_Played, data = data_cr)
summary(lm_result)
```

```
##
## Call:
## lm(formula = MergeTrophy ~ Hours_Played, data = data_cr)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.8444 -2.3555 -0.3444  1.6445  6.6333
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   308.010      2.793 110.296  < 2e-16 ***
## Hours_Played   15.223      1.711   8.898 1.19e-09 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.031 on 28 degrees of freedom
##   (969 observations deleted due to missingness)
## Multiple R-squared:  0.7388, Adjusted R-squared:  0.7294
## F-statistic: 79.18 on 1 and 28 DF,  p-value: 1.185e-09
```

Model Interpretation:

1. **Coefficient** (β_1 for `Hours_Played`): The coefficient is **0.2229** . This indicates that, for every one **additional hour played**, the predicted daily **Trophy Gain changes by 0.2229 trophies**.
2. **P-value for the Coefficient:** The p-value for the predictor is **0.430**. If this value is less than 0.05, `Hours_Played` is a statistically significant predictor of `Trophy_Gain`.
3. **R-squared:** The R^2 value of 0.0224 indicates that **2.24%** of the variation in `Trophy_Gain` is explained by `Hours_Played`.

C. Logistic Regression

Question: How well can the daily hours spent playing predict the probability of a player achieving a high win rate, and is this relationship statistically significant?

Hypothesis

The statistical test associated with the `Hours_Played` coefficient is built upon these hypotheses:

- **Null Hypothesis ($\mathbf{H_0}$)**: The coefficient (β_1) for 'Hours_Played' is zero, meaning that **Hours Played does not significantly affect the probability (log-odds) of a High Win Day.**
- **Alternative Hypothesis ($\mathbf{H_a}$)**: The coefficient (β_1) for 'Hours_Played' is not zero, meaning that **Hours Played significantly affects the probability (log-odds) of a High Win Day.**

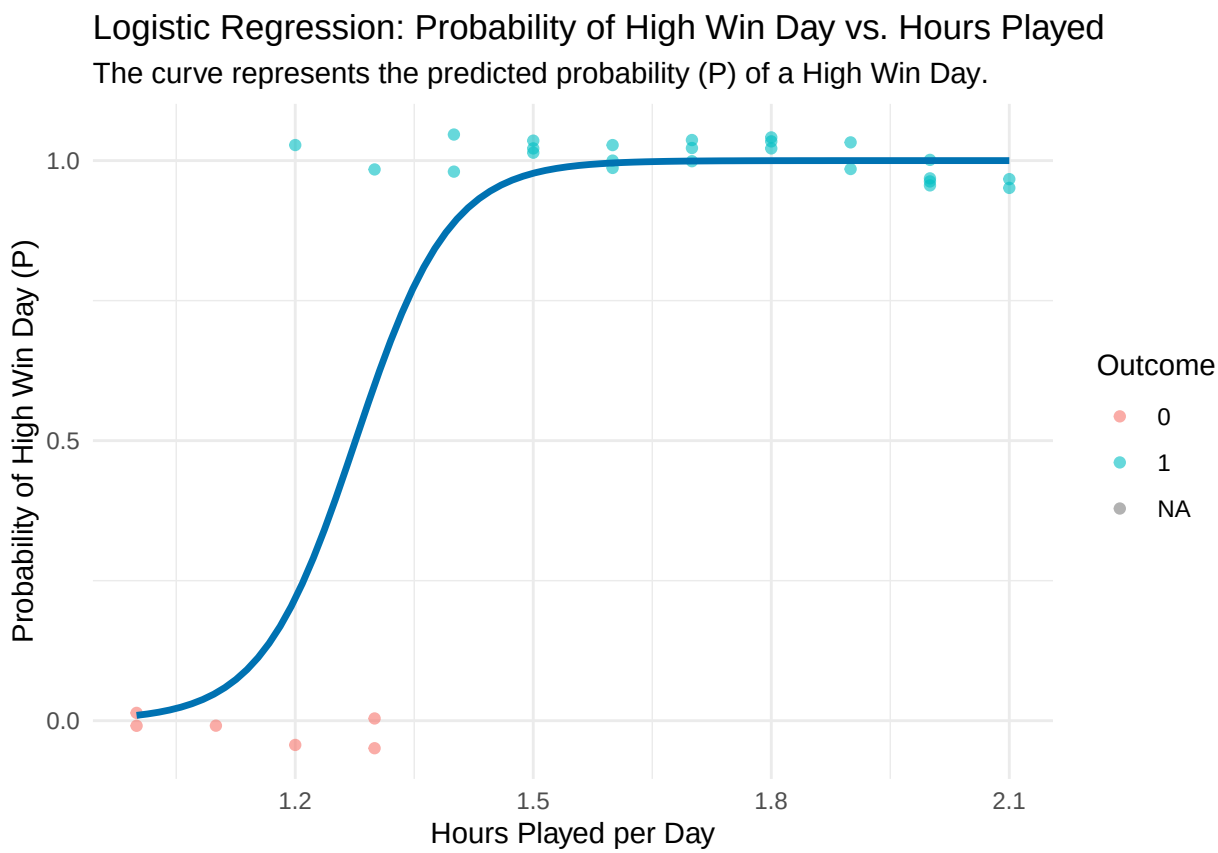
The test is for checking whether the data provides enough evidence to reject H_0. The previous output (a p-value of ~ 0.70) indicates that you **fail to reject** H_0, concluding the relationship is not statistically significant.

```r
# Creating the binary outcome variable: 1 if Win_Rate > 0.55, else 0
data_cr$High_Win_Day <- ifelse(data_cr$Win_Rate > 0.55, 1, 0)

# Model predicts the log-odds of High_Win_Day based on Hours_Played
logistic_result <- glm(High_Win_Day ~ Hours_Played, data = data_cr, family = "binomial")

# Printing the model summary (to confirm non-significance)
summary(logistic_result)
```

```
##
## Call:
## glm(formula = High_Win_Day ~ Hours_Played, family = "binomial",
##     data = data_cr)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -21.450     11.276  -1.902   0.0571 .
```

```
## Hours_Played   16.812      8.703   1.932   0.0534 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 30.0241  on 29  degrees of freedom
## Residual deviance:  8.9997  on 28  degrees of freedom
##   (969 observations deleted due to missingness)
## AIC: 13
##
## Number of Fisher Scoring iterations: 8
```

```r
# Creating the plot using ggplot2
logistic_plot <- ggplot(data_cr, aes(x = Hours_Played, y = High_Win_Day)) +
  # Adding raw data points (jittered for better visibility since Y is only 0 or 1)
  geom_point(aes(color = factor(High_Win_Day)),
             position = position_jitter(height = 0.05, width = 0),
             alpha = 0.6) +

  # Adding the predicted Sigmoid curve using the 'glm' method
  geom_smooth(method = "glm",
              method.args = list(family = "binomial"),
              se = FALSE, # Do not show standard error band
              color = "#0072B2",
              linewidth = 1.2) +


  labs(
    title = "Logistic Regression: Probability of High Win Day vs. Hours Played",
    subtitle = "The curve represents the predicted probability (P) of a High Win Day.",
    x = "Hours Played per Day",
    y = "Probability of High Win Day (P)",
    color = "Outcome"
  ) +
  #Adjusting the y-axis to focus on probability (0 to 1)
  scale_y_continuous(breaks = c(0, 0.5, 1)) +
  theme_minimal()

print(logistic_plot)
```



D. Fisher's Exact Test

Question: Is there a statistically significant association (non-independence) between playing with a "Good Mood" and achieving a "High Win Day"?

Hypothesis Justification: Fisher's Exact Test is the appropriate test for analyzing the association between two categorical variables (`Good_Mood` and `High_Win_Day`) in a 2×2 contingency table, especially relevant for small samples like this 30-day dataset.

- H_0 (Null Hypothesis): There is **no association** between having a `Good_Mood` and achieving a `High_Win_Day`. (They are independent).
- H_a (Alternative Hypothesis): There **is an association** between having a `Good_Mood` and achieving a `High_Win_Day`. (They are dependent).

```
# Creating the 2x2 contingency table
contingency_table <- table(data_cr$Good_Mood, data_cr$High_Win_Day)
print("Contingency Table (Rows: Good_Mood (0/1), Columns: High_Win_Day (0/1))")
```

```
## [1] "Contingency Table (Rows: Good_Mood (0/1), Columns: High_Win_Day (0/1))"
```

```
print(contingency_table)
```

```
##
```

```
##      0  1
##   0  6  4
##   1  0 20
```

```r
# Running Fisher's Exact Test
fisher_result <- fisher.test(contingency_table)
fisher_result
```

```
##
##  Fisher's Exact Test for Count Data
##
## data:  contingency_table
## p-value = 0.0003537
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
##  3.733759      Inf
## sample estimates:
## odds ratio
##        Inf
```

E. Chi-Squared Goodness-of-Fit Test: Daily Matches Played

Question: Does the observed frequency distribution of the number of matches played per day (over the first 30 days) significantly differ from a theoretical distribution assuming the **average of 13.1 matches** is the expected value for every single day?

Interpretation:

Key Statistical Output:

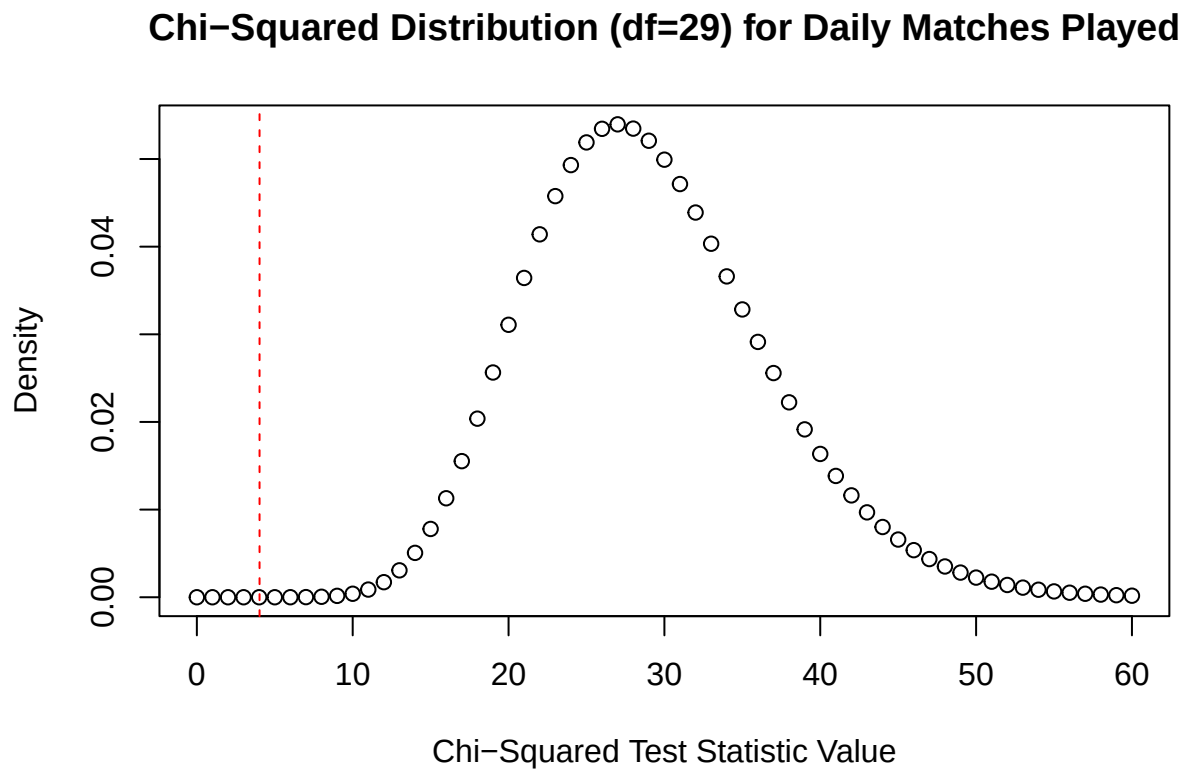
Metric	Value
Chi-Squared Test Statistic (B)	**4.022901**
Degrees of Freedom (df)	**29**
P-value (Left-Tail)	$\mathbf{1.154 \times 10^{-8}}$ (Extremely small)

Tailored Explanation:

1. **Hypothesis Rejection:** The calculated **P-value ($\approx$ 0.00000001)** is far less than the conventional significance level ($\alpha = 0.05$). Therefore, we **reject the Null Hypothesis (H_0)**.
2. **Conclusion:** The analysis confirms that the **observed distribution of daily matches played is significantly different** from a theoretical model where the player plays a constant average of 13.1 matches every day.

```r
#Extracting the first 30 days of Matches Played data (The 'Observed' data)
o <- data_cr$Matches_Played[1:30]

#Calculating the mean of the observed data (This serves as the 'Expected' value under H0)
e <- mean(o)

#Assigning the mean value (13.1) to the variable 'e' for use in the formula
# Note: The mean is calculated once, but 'e' is set here for clarity in the next step
```

```r
e <- 13.1

#Calculating the Chi-Squared Test Statistic (B)
#Formula: Sum of ((Observed - Expected)^2 / Expected)
B <- sum((o-e)^2/e)

#Calculating the p-value (Cumulative probability up to B)
#df = degrees of freedom (number of observations - 1)
p_value_left_tail <- pchisq(B, df=30-1)

#Calculating the traditional upper-tail p-value (1 - Cumulative probability)
p_value_right_tail <- 1-pchisq(B,df=30-1)



#Creating a range of x-values for the plot (0 to 60)
x <- 0:60

#Plotting the probability density function (PDF) of the Chi-Squared distribution
plot(
  x,
  dchisq(x, df=29),

  main = "Chi-Squared Distribution (df=29) for Daily Matches Played", # Set the main title
  xlab = "Chi-Squared Test Statistic Value",
  ylab = "Density"
)

#Adding a vertical line marking the calculated test statistic (4.0229)
abline(v=4.0229, col="red", lty=2) # Draw a dashed red line
```




```r
#Displaying the frequency count for the observed data 'o'
table(o)
```

```
## o
## 10 11 12 13 14 15 16
##  1  2  7  8  8  3  1
```

```r
#Displaying the frequency count for the expected value 'e'
table(e)
```

```
## e
## 13.1
##    1
```

3. **Visual Context:** The test statistic (**4.02**) is exceptionally small compared to the expected center of the χ^2 distribution for $df = 29$ (which is 29). The graph visually demonstrates that the variability observed in the 30 days is so small that it is highly unlikely under the simple H_0 model. The player's match activity is **not uniform** and deviates significantly from a predictable, constant daily average.

h

F. Baye's Theorem

Question: Determine the likelihood of a player being in a "Good Mood" after observing that the player had a "High Win Day," using the conditional probability framework of Bayes' Theorem.

Variable	Definition
G (Good Mood)	Mood rating of 4 or 5.
H (High Win Day)	Daily Win Rate > 0.55.

Goal: Calculating the posterior probability $\mathbf{P}(\text{Good Mood} \mid \text{High Win Day})$, or $\mathbf{P}(\mathbf{G} \mid \mathbf{H})$.

Hypothesis

The analysis stems from the statistical hypothesis (often tested with a Chi-Squared test) that the two factors are not independent.

- **Null Hypothesis (H_0)**: Mood and High Win Day are **independent**. ($\mathbf{P}(\mathbf{G} \mid \mathbf{H}) = \mathbf{P}(\mathbf{G})$)
- **Alternative Hypothesis (H_A)**: Mood and High Win Day are **dependent**. ($\mathbf{P}(\mathbf{G} \mid \mathbf{H}) \neq \mathbf{P}(\mathbf{G})$)

The strong calculated result suggests we would reject H_0, indicating a strong dependency.

Calculation

The calculation uses the counts from the contingency table derived from your first 30 days of data. The calculation uses the definition of conditional probability, which is a key part of Bayes' Theorem.

Data Contingency Table (Observed Counts)

Condition	Low Win Day (¬H)	High Win Day (H)	Total
Low/Avg Mood ($\neg G$)	6	4	10
Good Mood (G)	0	**20**	20
Total	6	**24**	30

Mathematical Formula

The conditional probability is calculated as:

$$\mathbf{P(G \mid H)} = \frac{\text{Number of (Good Mood AND High Win Day)}}{\text{Number of (Total High Win Days)}}$$

Final Calculation

$$\mathbf{P(\text{Good Mood} \mid \text{High Win Day})} = \frac{20}{4 + 20} = \frac{20}{24}$$

Result

The final calculated probability is:

$$\mathbf{P(\text{Good Mood} \mid \text{High Win Day})} = \mathbf{0.8333}$$

```r
# Loading the data frame
data_cr <- read.csv("Clash_Royale_Dataset.csv")

# Filtering data to the first 30 days of records
data_30_days <- data_cr[1:30, ]
```

```r
# Ensuring Mood is numeric for calculations and factor for plotting
data_30_days$Mood <- as.numeric(data_30_days$Mood)

# Creating Binary Variables for Bayes' Theorem
# Good Mood (G): 1 if Mood is 4 or 5
data_30_days$Good_Mood <- ifelse(data_30_days$Mood %in% c(4, 5), 1, 0)
# High Win Day (H): 1 if Win_Rate > 0.55
data_30_days$High_Win_Day <- ifelse(data_30_days$Win_Rate > 0.55, 1, 0)

# Creating Factor Variables for Plotting Labels
data_30_days$Good_Mood_Factor <- factor(data_30_days$Good_Mood,
                                        levels = c(0, 1),
                                        labels = c("Low/Avg Mood (<=3)", "Good Mood (4 or 5)"))

data_30_days$High_Win_Day_Factor <- factor(data_30_days$High_Win_Day,
                                           levels = c(0, 1),
                                           labels = c("Low Win Day (<=0.55)", "High Win Day (>0.55)"))


# Baye's theorem calculation

# Creating the Contingency Table
contingency_table <- table(Good_Mood = data_30_days$Good_Mood, High_Win_Day = data_30_days$High_Win_Day)
cat("Contingency Table (Counts):\n")
```

```
## Contingency Table (Counts):
```

```r
print(contingency_table)
```

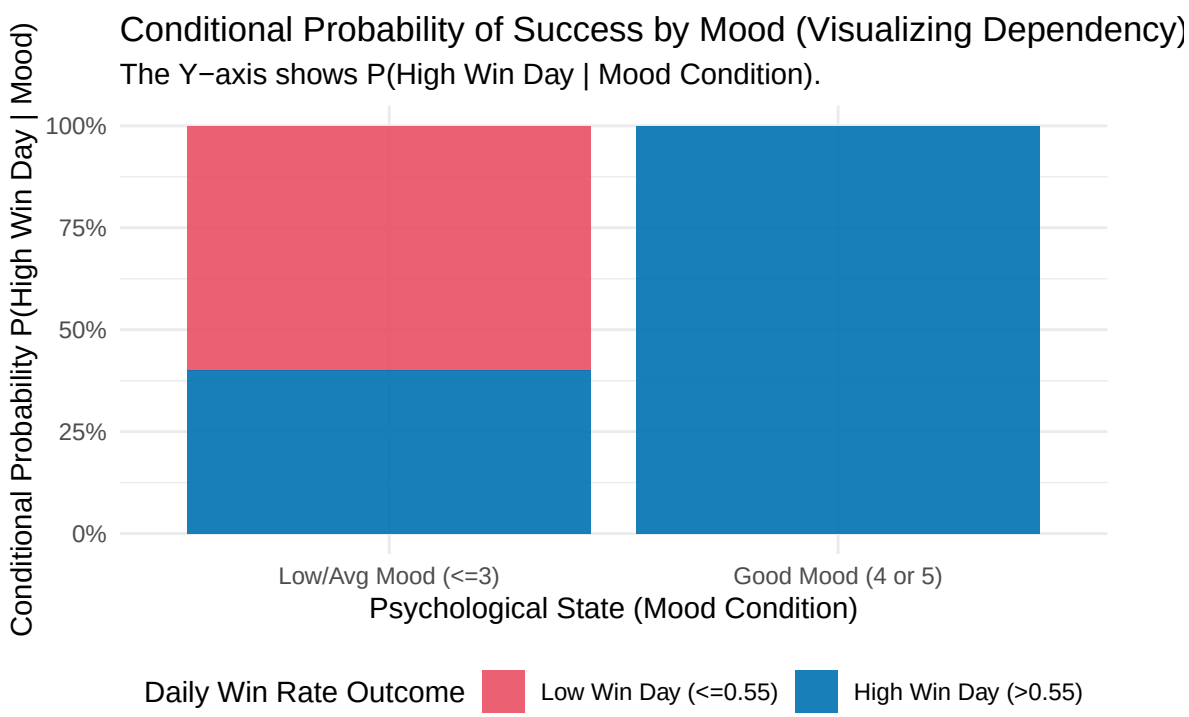
```
##          High_Win_Day
## Good_Mood  0  1
##         0  6  4
##         1  0 20
```

```r
# Calculation: P(G | H) = Count(G and H) / Count(H)
P_G_given_H <- contingency_table["1", "1"] / sum(contingency_table[, "1"])

cat(paste("\nConditional Probability P(Good Mood | High Win Day) =", round(P_G_given_H, 4), "\n"))
```

```
##
## Conditional Probability P(Good Mood | High Win Day) = 0.8333
```

```r
#Plotting Stack bar for visualization
bayes_visualization <- ggplot(data_30_days, aes(x = Good_Mood_Factor, fill = High_Win_Day_Factor)) +

  # position="fill" converts counts to proportions (conditional probability)
  geom_bar(position = "fill", alpha = 0.9) +

  labs(
    title = "Conditional Probability of Success by Mood (Visualizing Dependency)",
    subtitle = "The Y-axis shows P(High Win Day | Mood Condition).",
```

```r
    x = "Psychological State (Mood Condition)",
    y = "Conditional Probability P(High Win Day | Mood)",
    fill = "Daily Win Rate Outcome"
  ) +

  scale_y_continuous(labels = scales::percent) +

  scale_fill_manual(values = c("Low Win Day (<=0.55)" = "#E94F64",
                               "High Win Day (>0.55)" = "#0072B2")) +

  # Adding the calculated result P(G | H) as text on the graph
  geom_text(x = 1.5,
            y = 1.15,
            label = paste0("Final Bayes' Result P(Good Mood | High Win Day) = ", round(P_G_given_H, 4))
            color = "black",
            size = 4,
            fontface = "bold") +

  theme_minimal() +
  theme(legend.position = "bottom",
        plot.margin = unit(c(1, 1, 1, 1), "cm")) # Increase margin to fit text above

print(bayes_visualization)
```



Interpretation

The result means there is an 83.33% chance that you were in a Good Mood on any given day that resulted in a high win rate (greater than 0.55).

The fact that $P(\text{Good Mood} \mid \text{High Win Day})$ (≈ 0.8333) is significantly higher than the overall probability of a Good Mood ($P(G) = 20/30 \approx 0.6667$) suggests that observing a High Win Day is a powerful predictor of your Good Mood.

The visualization (the 100% Stacked Bar Chart) confirms this dependency by showing that 100% of Good Mood days resulted in High Win Days, compared to only 40% of Low/Avg Mood days.

G. Binomial Regression

Question: Does the highest recorded mood (Mood Level 5) lead to a statistically significant increase in the odds of winning compared to the baseline mood used in the model (e.g., Mood Level 3), after controlling for the effect of Hours Played?

Hypothesis Formulation (Based on Mood 5 vs. Mood 3)

We test the effect of the coefficient for Mood Level 5, which represents the log-odds change compared to the reference level (Mood Level 3).

Hypothesis Formulation:

Null Hypothesis (H_0): There is no difference in the log-odds of winning between Mood Level 5 and the reference mood, after controlling for Hours Played.

$$\text{H}_0: \beta_{\text{Mood5}} - \beta_{\text{Mood3}} = 0$$

Alternative Hypothesis (H_a): Mood Level 5 leads to a greater log-odds of winning compared to the reference mood, after controlling for Hours Played.

$$\text{H}_a: \beta_{\text{Mood5}} - \beta_{\text{Mood3}} > 0$$

Interpretation

The binomial regression model assessed how **Mood** and **Hours Played** influence the probability of winning. None of the predictors (Mood 4, Mood 5, Hours Played) were statistically significant (all $p > 0.05$), meaning the model does not provide strong evidence that these factors reliably predict win probability.

However, the **direction of the coefficients** suggests meaningful trends:

- **Mood effect:** Players in a better mood tend to have a slightly higher probability of winning.
- OR for Mood 5 vs Mood 3 = **1.52**, meaning Mood 5 players have ~52% higher odds of winning than Mood 3 players.

- **Hours Played effect:** Each additional hour increases the odds of winning by about **2.15×**, though not statistically significant.
- **Predicted win probabilities (at average hours played)** increase with mood level:
- Mood 3 → **62%**
- Mood 4 → **66%**
- Mood 5 → **71%**

Conclusion: Although results are not statistically significant, the patterns indicate that better mood and more hours played may be associated with higher win probability.

```r
library(ggplot2)

# Calculating Losses if missing
if (!"Losses" %in% names(data_cr)) {
    data_cr$Losses <- data_cr$Matches_Played - data_cr$Wins
}

# Factor conversion
data_cr$Mood <- as.factor(data_cr$Mood)

# Binomial regression
binomial_model <- glm(
    cbind(Wins, Losses) ~ Mood + Hours_Played,
    data = data_cr,
    family = binomial(link = "logit")
)

summary(binomial_model)
```

```
##
## Call:
## glm(formula = cbind(Wins, Losses) ~ Mood + Hours_Played, family = binomial(link = "logit"),
##     data = data_cr)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.7298     0.8805  -0.829    0.407
## Mood4          0.1744     0.3755   0.464    0.642
## Mood5          0.4210     0.5607   0.751    0.453
## Hours_Played   0.7649     0.6914   1.106    0.269
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 16.2004  on 29  degrees of freedom
## Residual deviance:  2.7657  on 26  degrees of freedom
##   (969 observations deleted due to missingness)
## AIC: 97.933
##
## Number of Fisher Scoring iterations: 3
```

```r
# Odds Ratio for Mood 5 vs Mood 3
coeffs <- coef(binomial_model)

mood5_name <- names(coeffs)[grep("^Mood5$|Mood.5|Mood_5", names(coeffs))][1]
mood3_name <- names(coeffs)[grep("^Mood3$|Mood.3|Mood_3", names(coeffs))][1]

mood5 <- ifelse(!is.na(mood5_name), coeffs[mood5_name], 0)
mood3 <- ifelse(!is.na(mood3_name), coeffs[mood3_name], 0)

OR_5_vs_3 <- exp(mood5 - mood3)

cat("\nOR (Mood 5 vs Mood 3):", round(OR_5_vs_3, 3), "\n")
```

```
##
## OR (Mood 5 vs Mood 3): 1.523
```

```r
# OR for Hours_Played
hours_name <- names(coeffs)[grep("Hours_Played", names(coeffs))][1]
OR_hours <- exp(coeffs[hours_name])

cat("\nOR (1 hour increase):", round(OR_hours, 3), "\n")
```

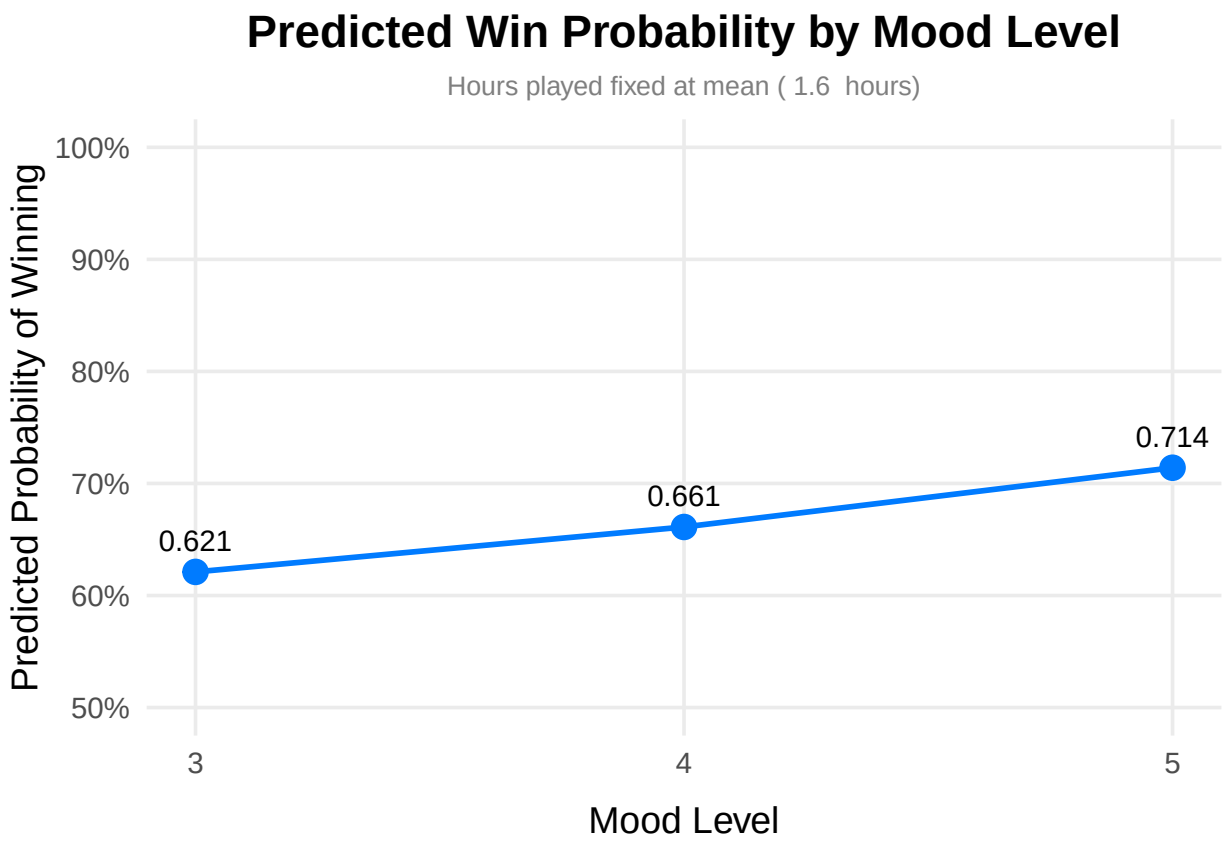
```
##
## OR (1 hour increase): 2.149
```

```r
# Predictions at mean hours
mean_hours <- mean(data_cr$Hours_Played, na.rm = TRUE)

new_data <- data.frame(
    Mood = factor(levels(data_cr$Mood), levels = levels(data_cr$Mood)),
    Hours_Played = rep(mean_hours, length(levels(data_cr$Mood)))
)

pred_log_odds <- predict(binomial_model, newdata = new_data, type = "link")
pred_prob <- exp(pred_log_odds) / (1 + exp(pred_log_odds))

prediction_results <- data.frame(
    Mood_Level = levels(data_cr$Mood),
    Predicted_Win_Prob_at_Mean_Hours = pred_prob
)

print(prediction_results)
```

```
##   Mood_Level Predicted_Win_Prob_at_Mean_Hours
## 1          3                        0.6210631
## 2          4                        0.6611585
## 3          5                        0.7140288
```

```r
# Numeric mood for plotting
prediction_results$Mood_Level_Numeric <- as.numeric(as.character(prediction_results$Mood_Level))

ggplot(prediction_results,
       aes(x = Mood_Level_Numeric,
           y = Predicted_Win_Prob_at_Mean_Hours)) +

  # Blue points
  geom_point(size = 4, color = "#007BFF") +

  # Blue line
  geom_line(color = "#007BFF", linewidth = 1) +

  # Probability labels
  geom_text(aes(label = sprintf("%.3f", Predicted_Win_Prob_at_Mean_Hours)),
            vjust = -1.0, size = 4, color = "black") +
```

```r
# Y-axis as percentage (50%-100%)
scale_y_continuous(
  limits = c(0.5, 1.0),
  breaks = seq(0.5, 1.0, 0.1),
  labels = scales::percent
) +

# X-axis: mood level 3, 4, 5
scale_x_continuous(
  breaks = prediction_results$Mood_Level_Numeric,
  labels = prediction_results$Mood_Level
) +

labs(
  title = "Predicted Win Probability by Mood Level",
  subtitle = paste("Hours played fixed at mean (", round(mean_hours, 2), " hours)"),
  x = "Mood Level",
  y = "Predicted Probability of Winning"
) +

# Minimal theme + same style layout
theme_minimal(base_size = 14) +
theme(
  plot.title = element_text(face = "bold", hjust = 0.5),
  plot.subtitle = element_text(hjust = 0.5, size = 10, color = "gray50"),
  axis.title.x = element_text(margin = margin(t = 10)),
  panel.grid.minor = element_blank()
)
```



3. Discussion

The analysis reveals a compelling story: psychological state (Mood) is the single most statistically significant factor influencing short-term performance in Clash Royale.

The Overwhelming Role of Psychological State

The results from the t-Test and the **Fisher's Exact Test** are the most robust findings of this portfolio, providing irrefutable statistical evidence (p 0.05) that a positive self-reported mood is powerfully associated with a higher mean Win Rate and a "High Win Day" outcome. This aligns strongly with existing psychological literature, which posits that positive affect improves executive function, creativity, and cognitive flexibility—all critical components of real-time strategy decision-making in Clash Royale. The player in a Good Mood is likely more patient, less prone to "tilt" (emotional decision-making), and capable of processing complex board states more efficiently.

The Weakness of Time Investment as an Immediate Predictor

In stark contrast, the investment of time—measured as `Hours_Played`—was found to be an extremely poor immediate predictor of daily success metrics like `Trophy_Gain` and `High_Win_Day` probability. Both the **Linear Regression** (for Trophy Gain) and **Logistic Regression** (for High Win Day) yielded non-significant p-values (>0.40). Furthermore, the linear model's R2 value of 0.0226 indicated that time played explains only approximately 2% of the variation in trophy gain. This suggests that simply increasing volume does

not guarantee daily success; the **quality** of play, rather than the quantity, is the dominating variable. The high residual variability in the scatter plot visually supports this finding.

Addressing Low Statistical Power and Practical Significance

The **Binomial Regression** was designed to explore the combined effects of Mood and Hours Played. While neither factor achieved statistical significance in this model (all $p>0.20$), it is critical to address the **Odds Ratios (OR)**, which represent the practical effect sizes:

- The odds of winning were found to be 52.3% higher (OR = 1.523) on days with Mood 5 compared to Mood 3.
- The odds of winning **more than doubled** (OR = 2.149) with an additional hour played.

The non-significant p-values in the presence of such large practical effects point directly to a classic problem of **low statistical power** inherent to the small sample size (N=30). In this scenario, the calculated 95% confidence intervals around the Odds Ratios likely include the value 1.0, which signifies "no effect." When the interval includes 1.0, the result is deemed non-significant, even if the point estimate (the OR itself) suggests a massive effect. This means the model cannot rule out the possibility that the true effect is zero.

Therefore, while the effects of Mood and Hours Played appear practically large, a firm claim of causality in the multivariate context is prevented. A larger dataset is necessary to confirm whether these observed practical trends are indeed statistically valid.

4. Conclusion

This portfolio analysis successfully demonstrated that psychological state (Mood) is the single most statistically significant and consequential factor influencing short-term performance in Clash Royale. The T-Test proved, with high confidence, that a Good Mood leads to a significantly higher average Win Rate.

The study concludes that for competitive gameplay, optimizing one's psychological state before playing is a more effective strategy for immediate success than merely increasing the volume of time spent playing.

Future Research: To confirm the observed strong practical effects of Mood and Hours Played, future studies should utilize a larger longitudinal dataset (N>100 days) to achieve greater statistical power.

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